

LESSON 128 (0.5 GROUND, 1.5 DUAL CROSS COUNTRY, and 0.3 INSTRUMENT)**Lesson Objective**

The purpose of this lesson is to introduce cross country flight procedures including navigation by means of pilotage and dead reckoning. The student will practice calculating ground speed and fuel burn by using actual time en-route (ATE) and comparing the results with his/her preflight planning.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Complete a cross-country plan for the route to be flown on 128. For example: KMLB-KOBE-KMLB or KMLB-KSUA-KMLB.
 - Review the material from lesson 127.
- Student questions.

References

- Advisory Circulars. ([Search Tool](#))
 - Aviation Weather. ([AC 00-6B](#))
 - Aviation Weather Services. ([AC 00-45H](#))
 - Pilot's Guide to a Preflight Briefing. ([AC 91-92](#))
- Aeronautical Information Manual.
 - Chapter 01 - Air Navigation.
 - Chapter 03 - Airspace.
 - Chapter 06 - Emergency Procedures.
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 16 - Navigation.

Lesson Content (Ground)**Review Cross-country Flight Planning**

- Airspace
 - Review the student's chosen route and verify the student's consideration regarding airspace.
 - Cruising altitudes should be appropriate for the airspace.
- Weight and Balance
 - Review the student's weight and balance planning on the TOLD card.
 - Verify that the fuel information on the TOLD card matches the fuel information on the navlog.
- Performance and Limitations
 - Review the student's calculations for pressure altitude and density altitude.
 - Verify that the V speeds were filled out correctly on the TOLD card.
 - Inspect the totals (times, fuel, distances).

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- **Navigation Log**
 - Verify the entirety of the nav log.
 - Cruising altitudes should be appropriate for the magnetic course.
 - Ensure that the airport diagrams are drawn, and the frequencies are filled in.
 - Verify that the student has the line drawn and checkpoints identified in the sectional chart.
 - Ensure that the student has a TOC and TOD checkpoint.
 - The information on the fuel management table must match the information on the front of the nav log.
 - Verify that the flight plan portion has the required and correct entries.
 - VOR information should be written.
- **Weather**
 - Review the student's weather planning.
 - Ensure that the student obtained a standard weather briefing.
 - Ask the student questions about any potential weather hazards for the route of flight.
 - Ensure that the student has filled out the weather section of the navlog.
 - Cruising altitudes should be appropriate for the weather.
- **Flight Plan Filed with FSS**
 - Ensure that the student has filed the flight plan with FSS.
 - Verify departure, enroute, and arrival times are correct.
- **Brief the student:**
 - Opening the flight plan.
 - Getting flight following.
 - Demonstrate on the first leg and let the student do it on the return leg.

Cockpit Management

- Brief management of radio frequencies.
- Advise how to keep resources orderly.
- Emphasize the need to think ahead of the airplane.

Visual Scanning and Collision Avoidance During High Workload**Situations**

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Runway Incursion Avoidance

- The student will be going to an unfamiliar airport. It's important to brief (using the airport diagram/google maps) the layout, taxiways, hold short line locations, etc. to avoid confusion and runway incursions.
- At this stage, the student should be familiar with runway incursion avoidance.
- Quiz if necessary.

Go/No-Go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Dual Cross Country)

Pilotage and Dead Reckoning

- Departure:
 - Set up radios, GPS, Nav aids, etc...
 - TIME OFF must be recorded.
 - Radio Communications:
 - Open flight plan
 - Receive flight following.
 - Coach student if needed.
- Cruise:
 - Make sure the student is switching tanks and leaning.
 - Evaluate CRM and division of attention.
 - Pilotage & Dead Reckoning
 - The student looks for checkpoints.
 - Course must be intercepted.
 - ATAs recorded.
 - Record actual Ground Speed.
 - Calculate revised ETA.
 - Verify location.
 - Complete the nav log.
 - Radio Communications
 - Coach student if needed.
- Arrival:
 - Use proper TOD.
 - Complete checklists.

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- Radio Communications:
 - Advise when to check the weather.
 - Cancel flight following.
 - Ask the student to close the flight plan.
- Completing Navigation Log
 - Complete last time box when safe.

Abnormal procedure

- Fail the MFD/PFD. Take away the moving map and all nav aids.
 - Pilotage & Dead Reckoning
 - ABNORMAL checklists complete.
 - PILOTAGE is the method of navigation.
 - Intercept course.
 - Record ATA.
 - Request revised groundspeed.
 - Calculate revised ETA.
 - Verify location.
 - Complete the nav log.
 - Radio Communications
 - Assist the student.

Emergency Procedure (Engine)

- Give the student a partial power loss/engine roughness.
- Evaluate their decision making.
- Ensure that the proper checklists are utilized.

Completing Navigation Log

- The nav log must be completed upon reaching the checkpoints.
- This includes:
 - Groundspeed.
 - Time/ATA.
 - Fuel burn.
- If the student is task saturated while crossing a checkpoint, remember:
 - Aviate, navigate, communicate!
 - Complete the nav log once time permits.

Basic Attitude Instrument Flying

- This lesson requires 0.3 instrument time.
- On the return leg, ask the student to put on their goggles/hood.
- The student must be able to perform:
 - Straight and level flight.
 - Climbs and descents.
 - Standard and medium level turns.

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- Intercepting and tracking a VOR and GPS course.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 129.
- Assign homework:
 - Review any deficiencies from the lesson.
 - Plan a cross country for lesson 129: KMLB-KSUA-KMLB, KMLB-KOBE-KMLB, or another suitable non-towered airport.

Completion Standards

This lesson is complete when the student is able to, before the flight (with minimal instructor guidance):

1. Have a completed navigation log prepared with minimal errors.
2. Have the correct climb, cruise and descent performance calculations prepared.
3. Obtain an official weather briefing and explain how the existing weather, NOTAMs and conditions will affect the flight.

This lesson is complete when the student, during the flight, is able to (with instructor guidance):

4. Intercept the planned course by using an on-course check point.
5. Have predetermined ETAs accurate to ± 7 min.
6. Calculate ground speed ± 10 kts based on actual time between checkpoints.
7. Maintain altitude ± 200 ft from desired altitude and $\pm 15^\circ$ from desired heading.
8. Use radio navigation, pilotage and dead reckoning procedures.

This lesson is complete when the student, during the flight, is able to (without instructor guidance):

9. Conduct all normal, abnormal and emergency checklist procedures.